



PATTERN AI LABS

EDITION · 2026 · FOR QSR

— INTRODUCING · EDGEVISION × FABRIC LENS

EdgeVision

Fabric Lens

The vision system that doesn't blink.

Custom-installed computer vision for quick-service restaurants — purpose-built cameras, on-site edge compute, and real-time intervention at the stations you choose. Machine oversight that is more accurate, more consistent, and never looks away.

• EDGE-FIRST • PRIVACY BY DESIGN • REAL-TIME INTERVENTION

EDGEVISION.PRO · PATTERN AI LABS

RETAIL · QSR
& DELIVERY HUBS

— THE PLATFORM · THE LENS

One edge platform. One QSR vision lens.

EdgeVision is Pattern AI Labs' on-prem Vision + GenAI edge runtime. **Fabric Lens** is the quick-service-restaurant product built on it — purpose-built industrial cameras, an on-site edge appliance, and signal fusion with the equipment already on your line. Start with one station, scale at your pace.

<300_{ms}

INFERENCE LATENCY · ON-SITE

Modular

ONE STATION OR ALL

24/7

CONTINUOUS WATCH

Real-time

DRIFT CAUGHT BEFORE HANDOFF

THE VISIBILITY GAP

Humans can't be every station, every second.

A QSR runs 8–12 stations across cook, prep, assembly and pass. A shift produces hundreds of orders, each carrying 15–30 quality and accuracy checkpoints. The math does not work for human oversight.

- **Attention fatigues** — human attention degrades within minutes; drift worsens as the shift wears on
- **Coverage is partial** — even the best line lead watches one station at a time — the rest run unobserved
- **Events are fast** — an assembly sequence takes seconds; a missing item a fraction of that — eye-blink fast
- **Audits are backward** — camera review happens after a complaint, by which point the cost has compounded



— THE COMPOUNDING COST

Every miss has a compounding cost.

The moments that drive refunds, NPS damage, complaints and brand erosion. None are visible in real time today. All can be detected and resolved in real time by Fabric Lens.

01 Burger count missed A double leaves the line as a single. The operator never sees it. REFUND BRAND TRUST SOCIAL COMPLAINT	02 Wrong sauce portioned The order spec says one sauce, a different one gets boxed. CUSTOMER RETURN ORDER REMAKE TICKET TIME	03 Modifier ignored Cheese on a no-cheese order. Allergen risk on the wrong customer. ALLERGEN RISK REFUND HEALTH COMPLAINT	04 Fry cook drift Fryer recovers slow on a rush. Pale fries reach the bag. NPS DROP REPEAT-RATE DECLINE REVIEW DAMAGE
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THE PRODUCT

A vision system that intervenes.

Fabric Lens is a modular, custom-installed computer-vision platform for QSRs. Industrial cameras at the stations you choose to instrument, an on-site edge appliance, and signal fusion with the equipment already on your line. Start with one station, scale at your pace.



01

Capture

Cameras at your priority stations



02

Detect

Task-specific models analyse every frame



03

Decide

Rules and fusion logic evaluate the scene



04

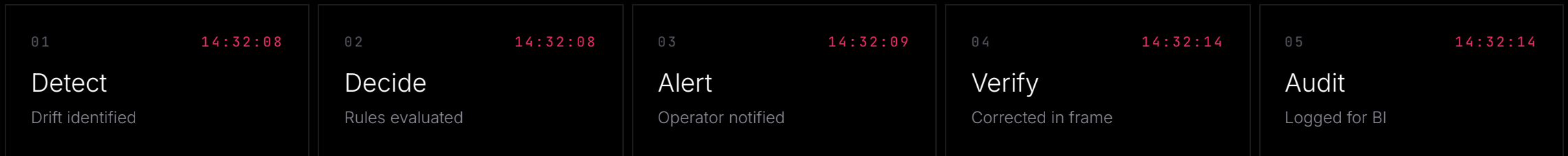
Act

Operator alerted at the station, in real time

— THE INTERVENTION LOOP

Detection is the start. Intervention is the point.

Real-time computer vision is only as valuable as the action it triggers. Fabric Lens closes the loop: every detected drift becomes an alert, an intervention, and an audit entry.

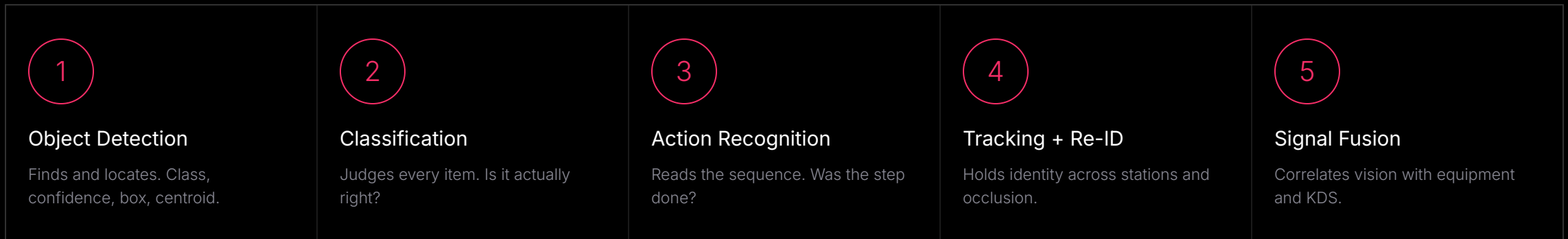


Example · missing-sauce intervention. 14:32:08 — order #4271 boxed; vision detects entrée and fries but no sauce cup, KDS calls for one. 14:32:09 — alert fires to the assembly screen, audio chime at pass. 14:32:14 — sauce added, re-verified, order released, event logged.

— HOW IT WORKS

Five techniques. One vision system.

A real CV product is not one model. It is a stack: detection finds what is there, classification judges if it is right, action recognition reads the sequence, tracking holds identity through occlusion, and fusion correlates all of it with equipment and order data.



— TECH 01-02 · PERCEPTION

Detection finds it. Classification judges it.

01 · OBJECT DETECTION

Detection finds it.

- Class label — one of the trained menu classes
- Confidence score — below threshold, the detection is suppressed
- Bounding box — the tightest rectangle that contains the object
- Centroid + scale — feeds tracking, counting and placement checks

02 · CLASSIFICATION

Classification judges it.

- ✓ Fine-grained, pixel-level quality assessment
- ✓ Answers the harder question: is the item right?
- ✓ Where a bounding box alone is not enough
- ✓ Catches the one item that fails (e.g. chickpeas fill 0.62 · LOW)

A box says something is there. Classification says exactly what — and exactly whether it is right.

TECH 03-04 · TIME & IDENTITY

Some quality questions only exist over time.

03 · ACTION RECOGNITION

Questions that only exist over time.

- A single frame cannot confirm a build sequence or a hygiene step
- Reads a sequence of frames, building confidence as motion completes
- Confidence ramps 0.12 → 0.97 as 'cheeseburger assembled' resolves
- Confirms the step actually happened, not just that items are present

04 · TRACKING + RE-ID

One order, followed end to end.

- ✓ Persistent identity on an order across stations and through occlusion
- ✓ Re-acquires the same order after it is briefly hidden
- ✓ ByteTrack and DeepSORT re-identification
- ✓ Enables true ticket-time decomposition — where the time was spent

Not just how long an order took — where it lost the time. Tracking plus KDS timestamps turn 'six minutes' into 'ninety seconds lost at assembly.'

— TECH 05 · THE MOAT

Vision + sensors + KDS. The layer no one else has.

Three input streams correlated frame by frame inside a single fusion core. It catches what one signal alone misses, separates equipment fault from human error, and confirms rather than merely detects.

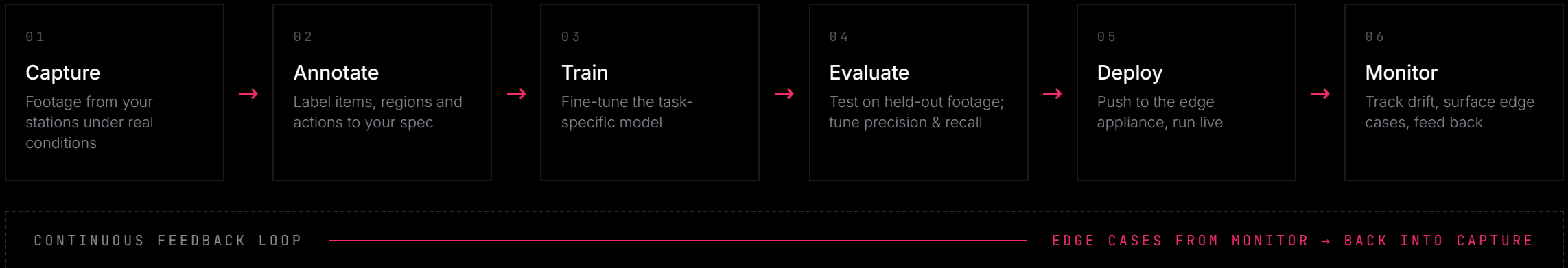
VISION Detection, classification, action recognition, tracking	EQUIPMENT TELEMETRY Fryer temp, cook timers, holding-cabinet state, weight sensors	KDS + POS Order content, modifiers, timestamps, bump events
SIGNAL FUSION CORE Frame-synced correlation across all three layers		

Fusion in one line. The fryer reports done-at-temp. Vision reports the product is under-colored. Neither signal alone catches it — together they say the equipment is drifting, not the cook.

CUSTOM MODELS

Data to deployment, then it never stops.

Every custom model moves through the same six-stage lifecycle. The first pass produces a working model; the loop keeps it accurate as your menu, lighting and seasons change.



— WHAT YOU CAN MEASURE

Six categories. Every one tied to the P&L.

Built from the way a multi-unit operator actually thinks. Each category ties to a revenue line or a brand-risk line, measured continuously through the fusion of video and equipment data.

01 Order Accuracy Perfect-order rate, item-level error, count accuracy, modifier compliance, condiment attach, packaging accuracy.	02 Food Quality & Consistency Cook doneness, cook-time conformance, temp at handoff, hold-time compliance, portion & build consistency.	03 Speed & Throughput Ticket time by station, station dwell, cook recovery, drive-thru time, throughput per labor hour, SLA breach.
04 Food Safety & Compliance Danger-zone events, hold-time violations, handwash compliance, cross-contamination risk, HACCP checkpoints.	05 Equipment Health & Calibration Calibration drift, oil quality & fryer health, recovery-time degradation, holding-cabinet conformance, sensor faults.	06 Labor & Waste Efficiency Staffing vs demand, throughput per labor hour, SOP adherence, overproduction waste, remake and refire rate.

— THE FUSION ADVANTAGE

The KPIs that only exist with fusion.

These five do not exist in a camera-only system or an equipment-only system. They require video and equipment data cross-referenced, frame by frame — the reason signal fusion is the moat.

01 Calibration drift Fryer reports done-at-temp; vision reports under-colored. The equipment is the problem, not the cook.	02 True ticket-time decomposition Not 'the order took six minutes' but 'it lost ninety seconds at assembly.'	03 Sensor-fault detection Holding cabinet reports a full pan; vision sees it empty. The sensor needs service — flagged.	04 Root-cause quality Oil TPM rising, product color darkening in step. The fries are worse because the oil is spent.	05 Predictive SLA breach Queue depth and per-station throughput forecast a ticket-time miss 60–90s before it happens.
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The moat. Competitors with cameras can copy detection. They cannot copy what they have not instrumented: the equipment signal layer underneath.

USE CASES · QSR

Speed, Accuracy, Safety — Live.

Three operational levers every QSR brand tracks weekly via spreadsheets and mystery-shopper audits. Fabric Lens tracks them per-shift, per-station, per-car — automatically.

SPEED OF SERVICE

— Drive-thru wait per lane, counter queue length, mobile-pickup dwell, time-to-first-bag — benchmarked per daypart against itself

KITCHEN CHOKE-POINTS

— Idle stations, expo bottleneck, fry-station coverage, grill hand-off delays — flagged the moment they exceed threshold

FOOD SAFETY & HYGIENE

— Gloves, hairnets, handwash frequency, holding-time at the heat lamp, spillage — scene-aware, low false-positive, audit-ready



USE CASES · EVERY SERVICE WINDOW

Every lane. Every car. Every second.

- Live per-lane queue length & wait time
- Order-board → handoff-window timing per car
- Pull-forward, blocked-lane & 'driver not served' alerts
- Lane-utilisation balance during peak hours
- Daypart vs forecast adherence



01

Counter & Kiosk

Queue vs labour by daypart · kiosk abandonment · greeter compliance · mobile-order handoff

02

Kitchen & Holding

Station coverage per shift · hot/cold-hold compliance · assembly time · wipe-down cadence

03

Cleanliness & Safety

Dining spillage & tray-bus cadence · restroom checks · slip-and-fall flagging · closing audit trail

04

Compliance & Audit

PPE per station · handwash vs SOP · camera-health monitoring · executive summary email

— PRODUCT ARCHITECTURE

Cameras, compute, sensors — edge-first, cloud-managed.

Three zones: a custom install in the restaurant, one edge appliance running vision and fusion locally, and AWS-native managed services for training, fleet management and reporting.

01 On-premise CUSTOM INSTALL PER STORE • Fryer cam + temp probe • Cook line profile + top vision • Assembly top-down + weight sensor • Pass / handoff + KDS feed	02 Edge appliance · Lens ONE PER RESTAURANT • Stream ingest + detection models • Tracking + Re-ID • Sensor fusion + rules engine • Event generator · Intel Core Ultra ~100 TOPS	03 AWS cloud MANAGED SERVICES • Model training · SageMaker • Foundation models · Bedrock • Event bus · EventBridge + IoT Core • Storage · S3 + Timestream
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259_{ms}

INFERENCE P50

4–16

4K · 30 FPS STREAMS

ZERO

CLOUD ROUND-TRIP IN DETECT PATH

us-east-1

MULTI-AZ

HARDWARE & PRIVACY

Intel-powered. Privacy by design.

High-performance Intel edge compute on the site LAN reads existing and purpose-built cameras, and runs the full detection-and-fusion pipeline locally. The cloud is for training and fleet management — never the real-time detection path.

HARDWARE · REFERENCE EDGE NODE

PLATFORM	Lenovo · Model TBD
PROCESSOR	Intel Core Ultra 9 285H · ~100 TOPS
MEMORY	32 GB
STORAGE	1 TB NVMe
OS	Ubuntu 24.04 LTS
NETWORK	Closed LAN, isolated

PRIVACY & SECURITY

- ✓ On-prem inference — detection & fusion run inside the site
- ✓ No frames or biometrics leave in the detection path
- ✓ Role-based access (RBAC) for managers & HQ
- ✓ FIFO video retention with configurable policy
- ✓ Tamper-resistant edge appliance
- ✓ Zero cloud round-trip for real-time alerts

THE PARADIGM SHIFT

Why fusion + GenAI changes everything.

DIMENSION	TRADITIONAL CV	EDGEVISION FABRIC LENS
Detection	Bounding boxes & pixel thresholds	Contextual scene understanding
Accuracy	High false-positive in varying light	Adaptive to lighting & occlusion
Quality	Cannot judge if the item is right	Fine-grained, pixel-level classification
Root cause	Counts and timestamps only	Fusion separates equipment from human error
Adaptability	Manual re-calibration per store	Continuous-learning custom models
Insights	Tells you what happened	Tells you why — and intervenes in real time

“Traditional analytics tells you an order left the line. Fabric Lens tells you the double left as a single, the sauce cup was missing, and the fryer — not the cook — was drifting.”

— PRODUCTION LINEAGE

This team has shipped production CV before.

Fabric Lens is not a first attempt. The custom-model approach, the edge architecture and the QSR domain work all trace to production deployments the team has built and run.

01

BigBasket Fresho

Custom vision models for produce quality and self-checkout. Retrained per category, in production at scale.

800+ SKUS LIVE

<300MS INFERENCE

02

Wonder · QSR

Quality and consistency analytics purpose-built for a multi-unit QSR operator. Vision plus operational signal.

MULTI-UNIT

QSR DOMAIN

03

Pattern AI · EdgeVision

Edge-first computer vision with on-site inference and real-time event generation — the architectural lineage for Lens.

EDGE-FIRST

REAL-TIME EVENTS

— ENGAGEMENT

Pilot to production. Then chain rollout.

Custom install means a structured roadmap. Each phase has clear outputs and acceptance criteria; pricing is tiered by chain size to support both pilot economics and scale economics.

PHASE 00 Week 0-1 Discovery Camera audit, station prioritisation, AWS architecture review, pilot-store selection.	PHASE 01 Week 2-6 Install + pilot Custom camera install, edge appliance, first signature model, UAT on the pilot station.	PHASE 02 Week 7-14 Hardening Additional models, sensor fusion, dashboard tuning, multi-store readiness.	PHASE 03 Week 15+ Chain rollout Zero-touch provisioning, per-store onboarding, continuous model expansion.
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What you get at the end of the pilot: a running Fabric Lens install on your priority stations, a first signature model on your own footage, a live intervention loop, and a dashboard your operators have run a full daily cycle on.



See every station. Catch every drift. Intervene in real time.

Production-grade computer vision for QSR operations. Custom models, custom-installed, edge-first. Methodology you can audit, KPIs you can run the business on.

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